1. In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

Cisco Packet Tracer: DHCP Server with 2 PCs

1. Add devices

In Packet Tracer, add:

- **1 Server** → DHCP Server
- **1 Switch** → 2960
- **2 PCs** → PC0 and PC1

Connect them using **Copper Straight-Through** cables:

```
PC0 ────────┐
   │
PC1 ────────┼── Switch ── DHCP Server
   │
```

2. Configure the DHCP Server

Click **Server → Desktop → IP Configuration**.

Set:

IP Address: 192.168.10.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

Then go to:

Server → Services → DHCP

Turn DHCP **ON** and create a pool:

Pool Name: LAB_POOL

Default Gateway: 192.168.10.1

DNS Server: 8.8.8.8

Start IP Address: 192.168.10.100

Subnet Mask: 255.255.255.0

Maximum Users: 50

Click **Add/Save**.

3. Configure PC0 for DHCP

Go to:

PC0 → Desktop → IP Configuration → DHCP

PC0 should automatically receive an address such as:

IP Address: 192.168.10.100

Subnet Mask: 255.255.255.0

4. Configure PC1 for DHCP

Go to:

PC1 → Desktop → IP Configuration → DHCP

It should receive a different address, for example:

IP Address: 192.168.10.101

Subnet Mask: 255.255.255.0

5. Verify using Command Prompt

On **PC0**:

- ipconfig

On **PC1**:

- ipconfig

You can also test connectivity:

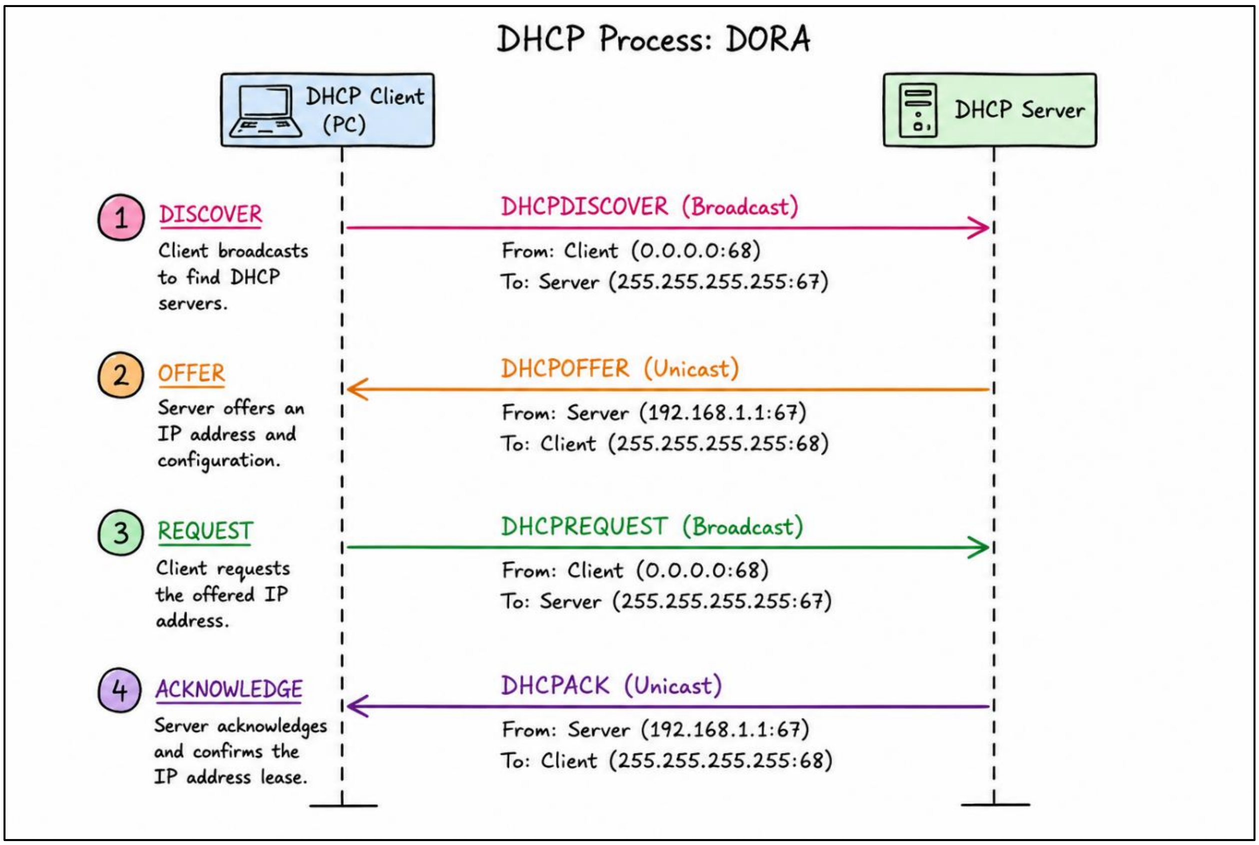
- ping 192.168.10.101

from PC0.

Expected Result

Device	IP Address	Source
DHCP Server	192.168.10.2	Manually configured
PC0	192.168.10.100	DHCP
PC1	192.168.10.101	DHCP

2. Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage.



3. Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.

➢ Only **one IP address (192.168.10.100)** is available in the DHCP pool. The first PC receives and leases this address, leaving no available address for the second PC. Therefore, the second PC cannot obtain an IP address automatically.

4. Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.

Cisco Packet Tracer: Zomato-Style Restaurant Wi-Fi with DHCP

1. Create the topology

Add these devices:

- **1 DHCP Server**
- **1 Wireless Access Point**
- **1 Laptop**
- **1 Generic PC** → represents the smartphone

Connect:

2. Configure the DHCP Server

Go to:

Server → Desktop → IP Configuration

Set:

IP Address: 192.168.10.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

Then:

Server → Services → DHCP → ON

Create this pool:

Pool Name: RESTAURANT_WIFI

Default Gateway: 192.168.10.1

DNS Server: 8.8.8.8

Start IP Address: 192.168.10.100

Subnet Mask: 255.255.255.0

Maximum Users: 50

Click **Add**.

3. Configure the Wireless AP

Set the wireless network name (SSID) to:

ZomatoWiFi

For a simple Packet Tracer simulation, connect the AP to the DHCP server using Ethernet.

4. Configure the Laptop

Connect the laptop to **ZomatoWiFi**, then select:

Laptop → Desktop → IP Configuration → DHCP

Expected result:

IP Address: 192.168.10.100

Subnet Mask: 255.255.255.0

5. Configure the Generic PC as Smartphone

Connect the Generic PC to the wireless network and select:

PC → Desktop → IP Configuration → DHCP

Expected result:

IP Address: 192.168.10.101

Subnet Mask: 255.255.255.0

6. Verify the assigned IPs

On each device, open **Command Prompt** and run:

- Ipconfig

Device	Role	DHCP IP
Laptop	Restaurant staff/customer device	192.168.10.100
Generic PC	Smartphone	192.168.10.101
DHCP Server	Network server	192.168.10.2